Single cycle processor Data path & execution (Note)

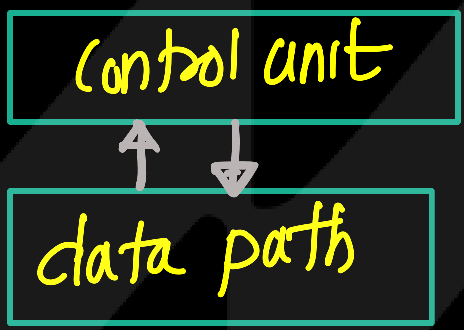
→ Single cycle processor execute one instruction in one clock cycle.

→ Designing of single cycle processor

① Constructing the data path by connecting the state elements and combinational logic circuits.

② Construct control unit using combinational logic circuit that generate appropriate control signal.

③ Analyze the performance of single cycle processor



Single cycle processor — sample program - ①

① LW x_5, 4(x_9)

→ x_5, x_9 are the register in Register file

Content of x_9 = 2000 H →

→ Content of PC = 1000 → address of instruction stored at 1000 in the instruction memory.

→ Content of memory location 2004 = '6'

→ LW x_5, 4(x_9)

load the content of memory location pointed by 4(x_9) to the register x_5

effective address of memory location can be determined by adding base Address with offset address

① base add is present in $x_9 \Rightarrow 2000$

② offset $= 4$

effective address $= 2000 + 4 = 2004$

→ Content of 2004 is moved to x_5.

→ LW x_5, $4(x_9)$ is a I-type instruction

12	5		5	7bit
imm (4)	rs1 (x_9)	funct 3	rd (x_5)	op
31 — 20	19 — 15	(14 — 12)	(11 — 7)	(6 — 0)

12	5		5	7bit
0000 0000 0100	0 1001	010	00101	0000011
31 — 20	19 — 15	(14 — 12)	(11 — 7)	(6 — 0)

Instn 6:0 → represents opcode

Instn 11:7 → 〃 destination

instn 14 : 12 — funct 3 bits

instn 19 : 15 — source

instn 31 : 20 → represent immediate value

base address (x_9) represented by instn 19 : 15 bits

offset (4) represented by instn 31 : 20 bits.

Step 1 – Construct data path.

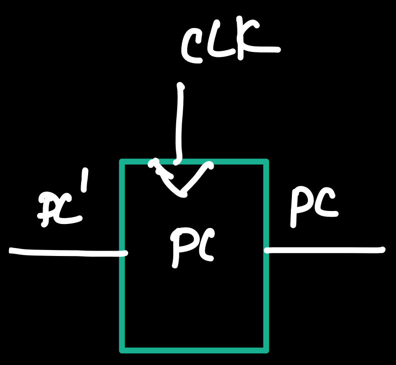
① state elements required

① PC, Instruction memory, Register file, data memory

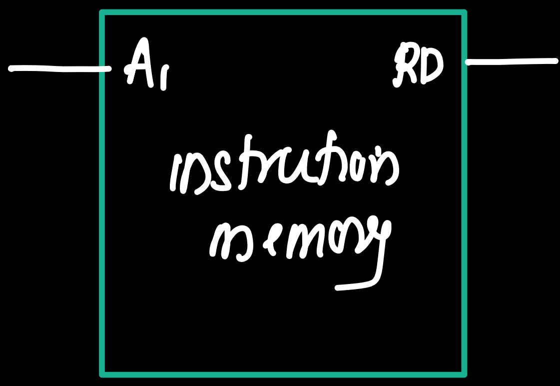
② Additional logic circuit

① ALU

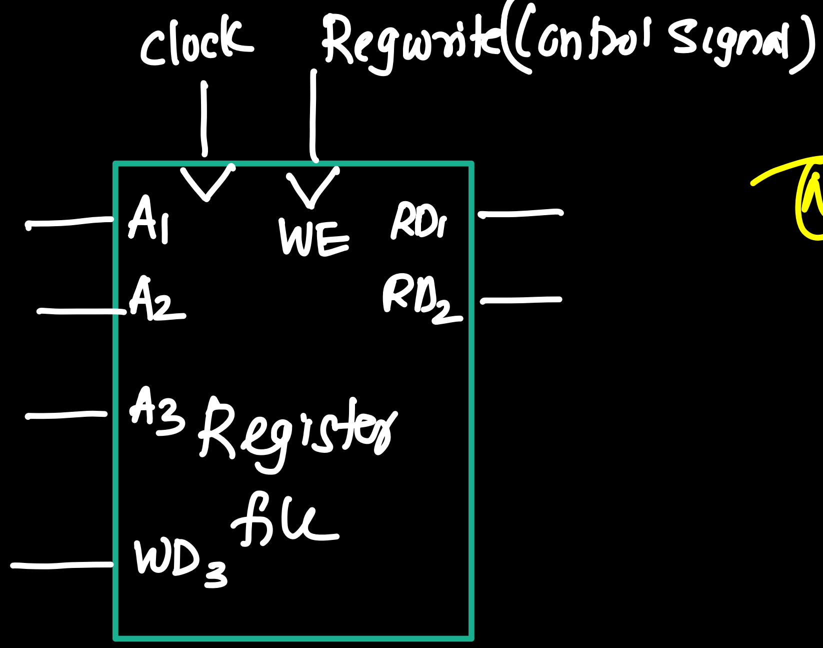
State elements

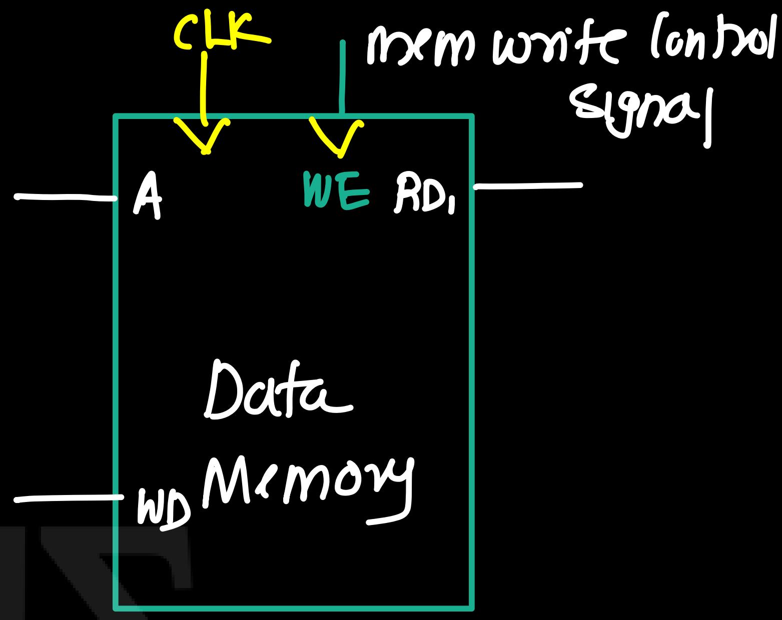


No control signal



No clock
no control signal



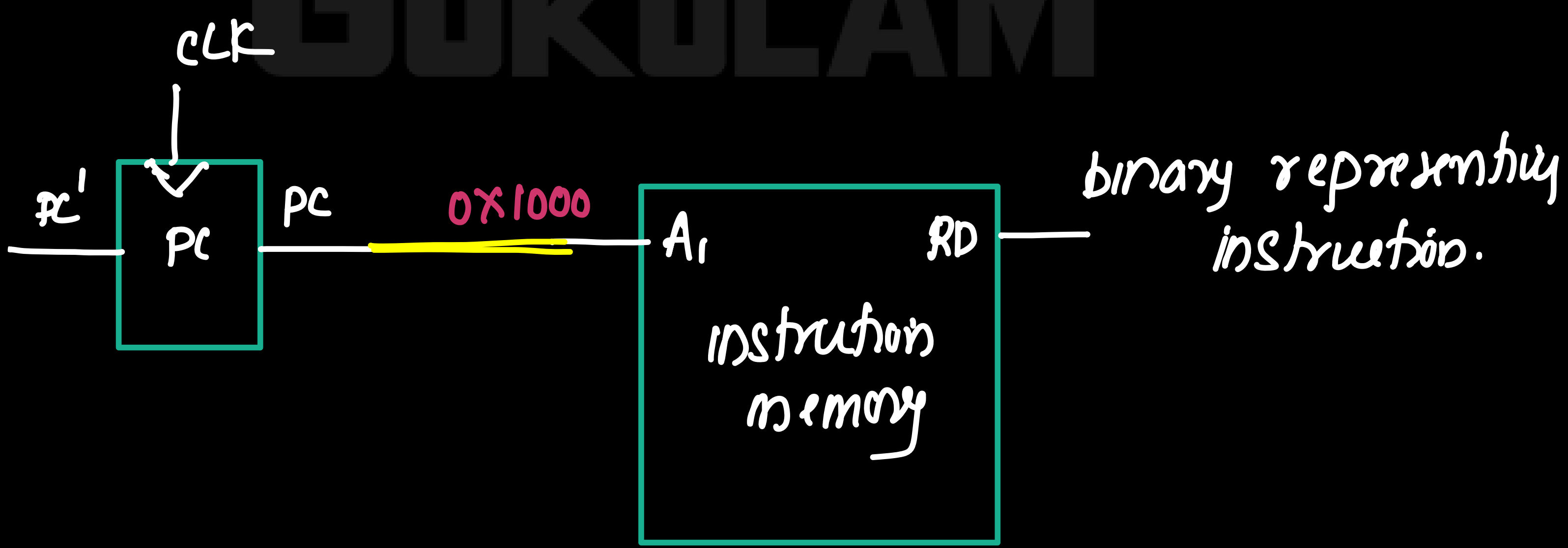


Instruction Execution Steps

① PC Contains address of next instruction to be executed

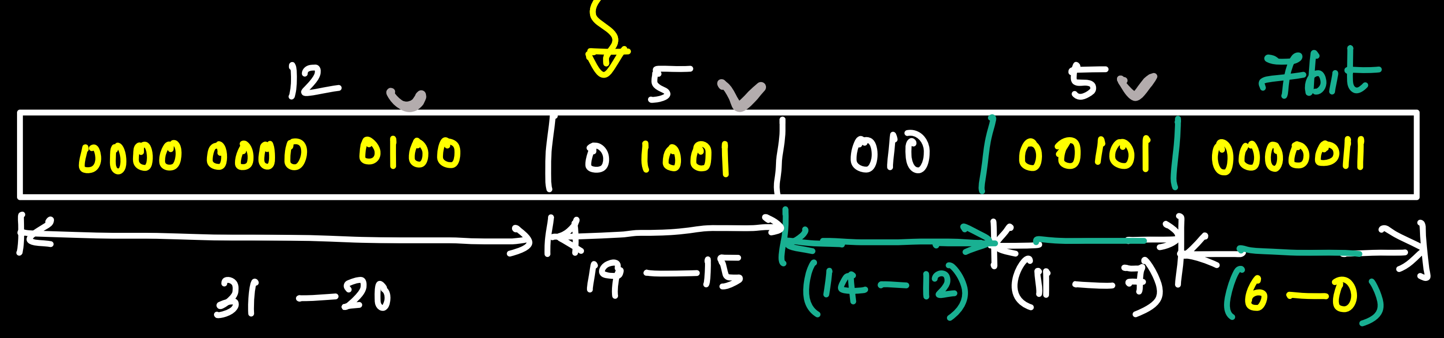
$PC = 1000H$

② PC output is directly connected to the address input of instruction memory.



② instruction memory fetch the instruction whose address is present at the address input. and output the instruction through read data output (RD1). thus instruction fetch is completed

$RD_1 \rightarrow$ Contains machine code (binary) of instruction



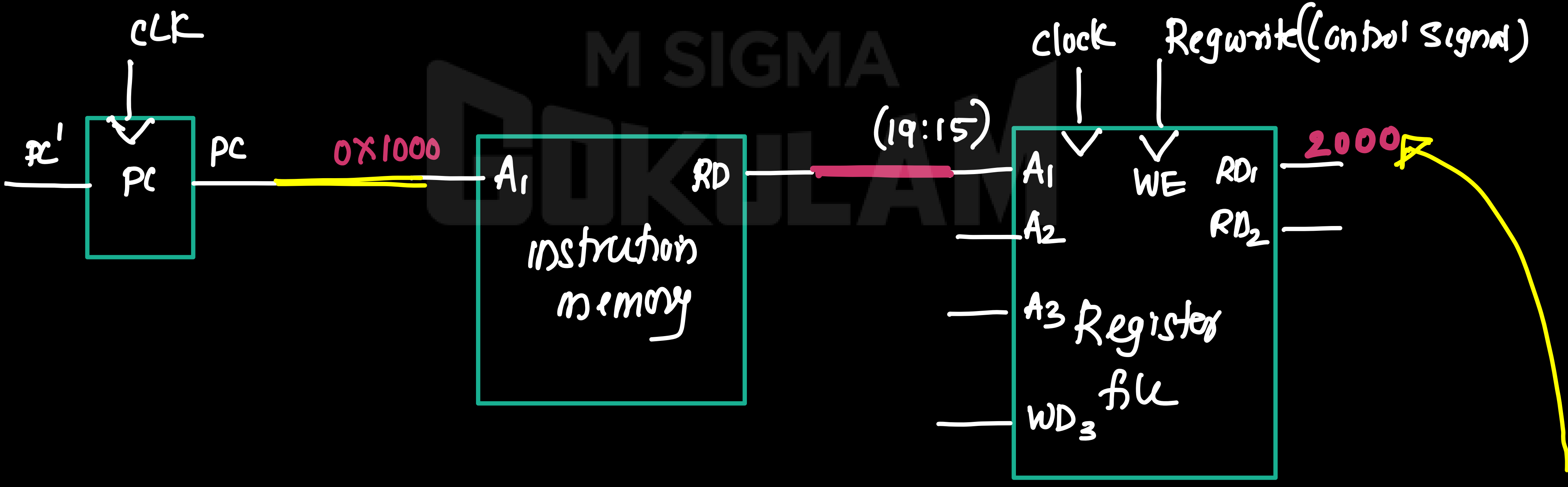
Instn 6:0 → represents opcode
Instn 11:7 → " destination
instn 14:12 — funct 3 bits
instn 19:15 — source
instn 31:20 → represent immediate value

base address (xa) represented by instn 19:15 bits
offset (4) represented by instn 31:20 bits.

③ Next is to calculate the effective address of memory location.

— read the source register (x9) containing base address.

— instruction bits 19:15 (instn 19:15) represents the source register. So connect these bits to the address input (A_1) of Register file.

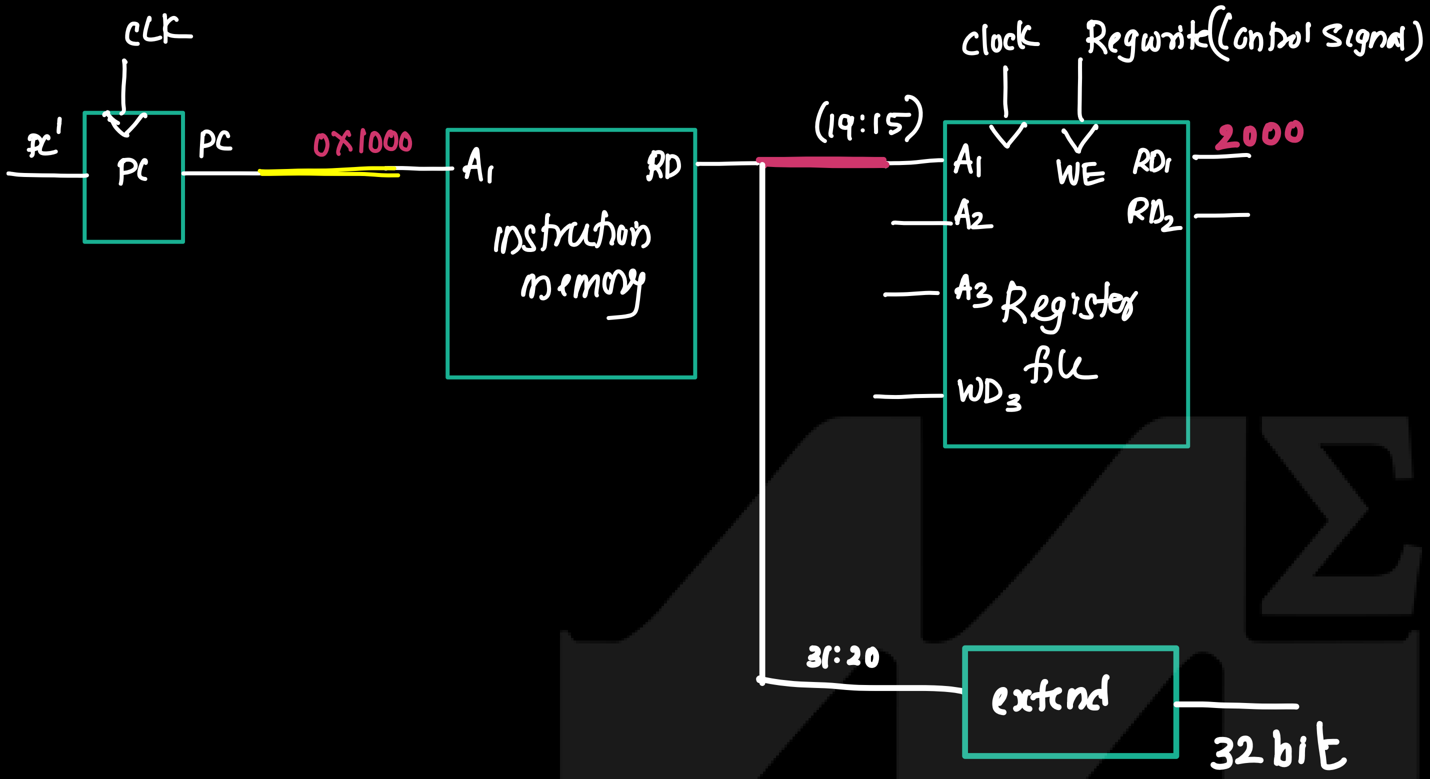


— Register file Read the value of register specified by the address in A_1 and output to the RD1 (RD1 = content of x9 = 2000H)

— To calculate effective address, need offset.
· offset value stored in the instruction (bits 20 to 31) (instn 31:20) → 12 bit.

→ To calculate effective address, need to add base address (2000) and offset 4 (12bit)

· to add these two extend 12 bit (offset) to 32bit. this can be done by passing 12 bits of instruction (instn 31:20) to the extender.

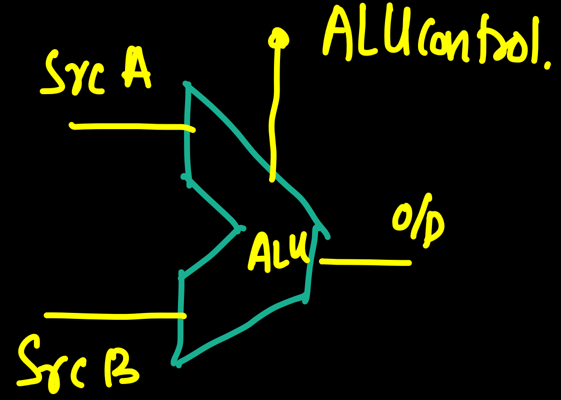


→ processor adds base address (RD1) and offset (32bit) to get the effective address. This addition can be done by ALU.

— ALU receives two input (Src A, Src B). Src A is the base address from the register file and Src B is the offset from the 'extend'.

→ ALU can perform many operations. (Add, sub etc..) So a control signal is required to specify the operation (ALU control).

If ALU Control → 000 → then, perform addition

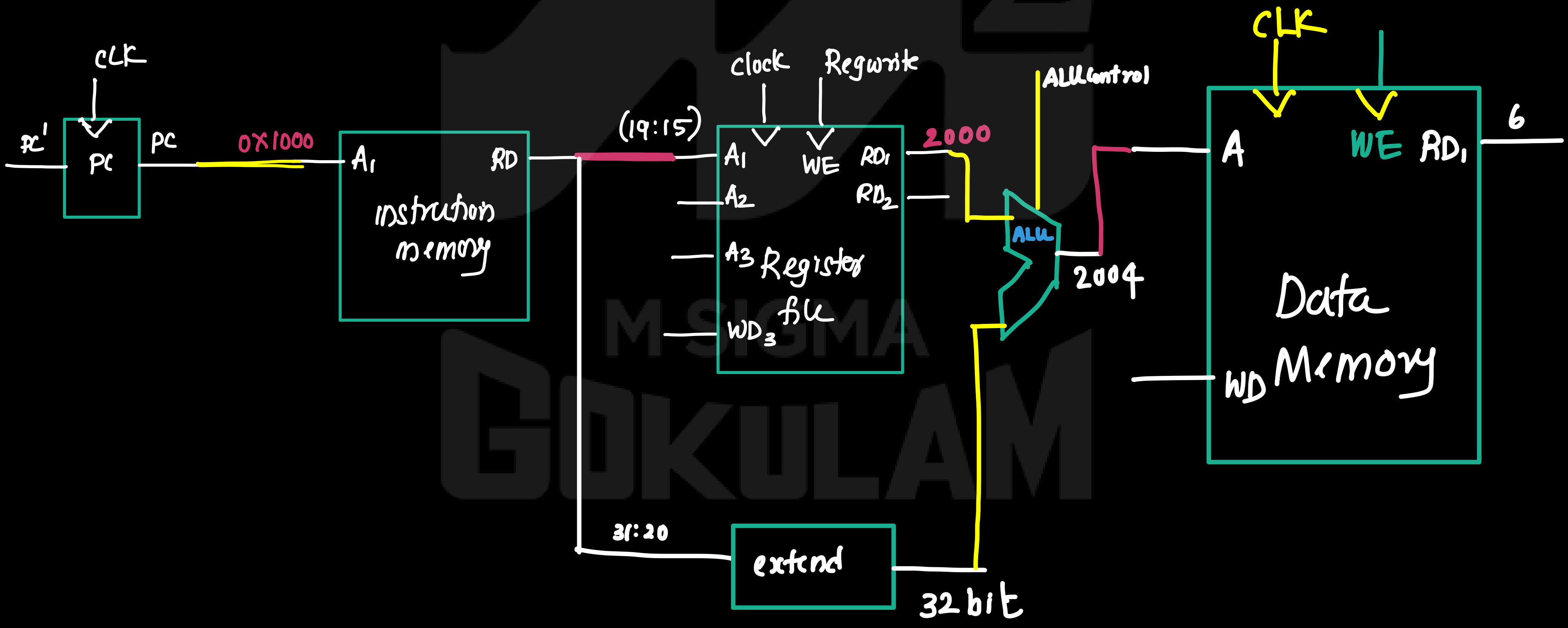


→ ALU produce the effective address.
→ output of ALU is send to the address input

of data memory.

→ In data memory, two operations are possible
① read ② write
for selecting the operation a control signal should be given to data memory. That control signal is called memory write (Memwrite). this is connected to write enable (WE) pin
If WE = 1, then perform write operation
If WE = 0, then perform read operation.

→ Data corresponding to address present in the read input is placed to the read data output (RD_1).



→ The data available at RD_1 need to written in the register (x_5). Instruction bits 7 to 11 (instn 11:7) represent the destination register. so these bits (instn 11:7) are connected to A_3 of Register file. and RD of Data memory connected to WD_3 of Register files.

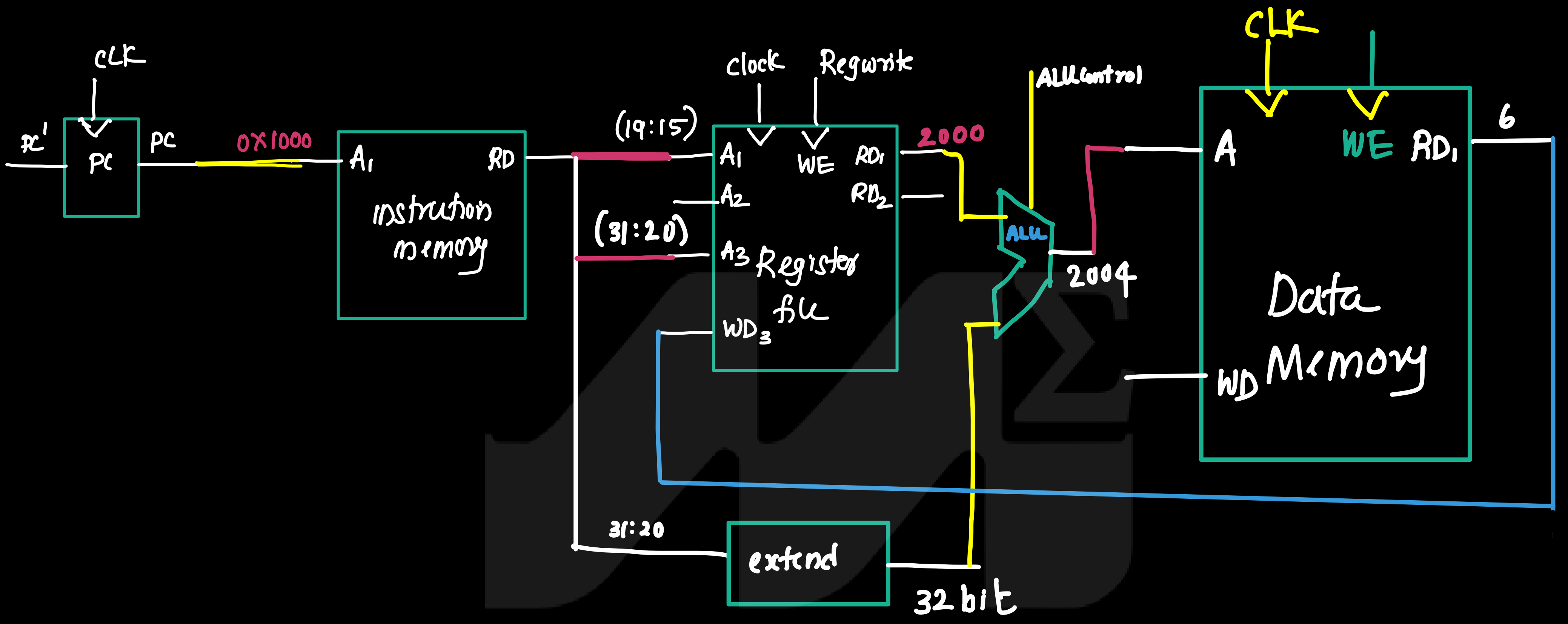
→ two operations can be performed on register files (Read & write). operations can be selected by giving a control signal (Reg write).
Register write.

If Regwrite = 1 → write operation
If Regwrite = 0 → read operation.

→ If control signal (Regwrite) at WE pin is one then write on the memory data available at WD_3 is location whose address is available at A_3.

→ Finally data is written on the register file.

→

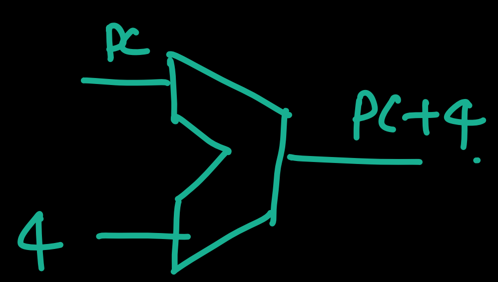


→ While executing one instruction, processor must compute the address of next instruction (PCNext)

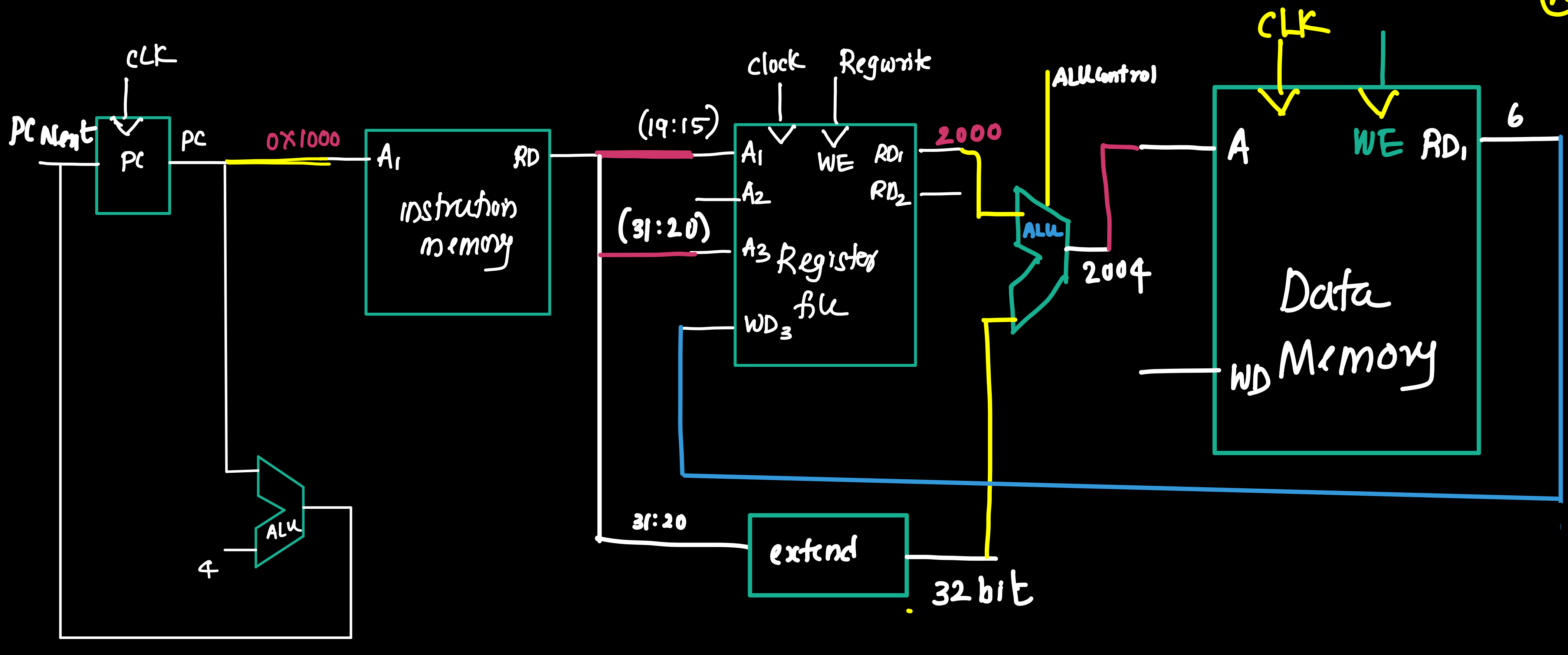
$$PC_{next} = PC + 4$$

Instruction are 32 bit (4 byte). so address of the next instruction is PC+4.

→ Use an adder to find PC next and result is written on the 'PC'.



N



NC

Summary: LW x5, 4(x9)

① PC provides address to the instruction Memory. which output the instruction

② Register file reads (x9). and extend unit create 32 bit extended immediate value.

③ ALU adds the extended immediate value and base address to calculate the memory address.

④ Data memory read the value from the address specified by the ALU output.

⑤ Store the data to the destination register (x5)

⑥ Update the PC to determine the address of next instruction

$PC_{Next} \longrightarrow \underline{\underline{PC+4}}$

Single cycle RISC processor Datapath – Example-2

(N)

① Instruction : sw x_5, 4(x_9)

- store the content of memory location to the register (x_5).

effective address = base address + offset

$\downarrow$

effective address = Content of x_9 + 4

- Content of x_9 = 2000
- Content of memory location 2004 = 06.
- PC = 1000.

sw x_5, 4(x_9) — x_5 is rs_2, x_9 is rs_1

'sw' is 's' type instruction

7	5	5	3	5	7
imme	rs_2 (x_5)	rs_1 (x_9)	f3 (010)	imm (00100)	OP
0000 000	00101	01001			
(31 : 25)	(24 : 20)	(19 : 15)	(14 : 12)	(11 : 7)	(6 : 0)

In this instruction immediate value represented by (instn 11:7, instn 31:25).

① PC place the address to address input (A1) of instruction memory.

② instruction memory fetch the instruction and place the instruction in read data output (RD).

③ Address calculation :

→ To calculate the effective address, need to read the content of register (x_9) and extend the immedia. value represented by instn 11:7, instn 31:25

→ To read the content of register (x_9), instruction bits (instn 19:15) is connected to address input of register file

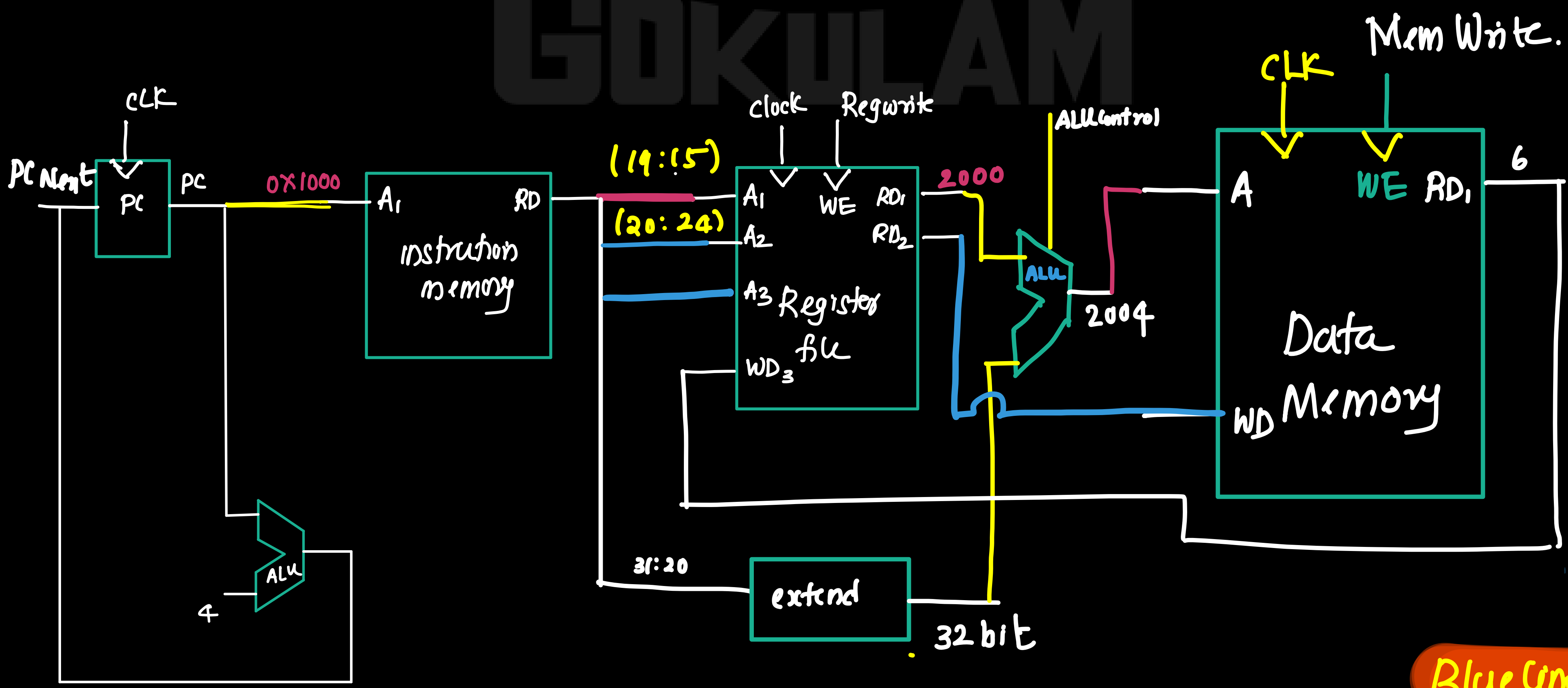
→ register file read the register (x9) and place the content to read data output (RD1).

→ immediate value represented in the instruction bits (instn 11:7, instn 31:25) is given to extender. and apply control signal (Imm Src = 1). then extender produce 32 bit immediate value.

— base address of the RD1 and 32 bit immediate value given to the ALU. ALU add these values and produce effective address. output of ALU is connected to the address input of data memory.

— to read the content of x_5, connect instruction bit (instn 20:24) to the address input (A_2) of register file. Register file read the content and place in RD2. Connect this RD_2 to the WD pin of Data memory.

→ when control signal (MemWrite) is enabled (WE=1), then data at the WD_3 pin is stored at memory location whose address is specified in the address input (A_3).



Blue lines are New Line